

1) a) let $\sigma(x) = \frac{1}{1+e^{-x}}$ show that $\sigma'(x) = \sigma(x)[1-\sigma(x)]$

$$\frac{d\sigma}{dx} \left[\sigma(x) = \frac{1}{1+e^{-x}} \right]$$

$$\frac{d\sigma}{dx} = (1+e^{-x})^{-1} \cdot dx \quad u(x) = 1+e^{-x}$$

$$\frac{d\sigma}{dx} = -(1+e^{-x})^{-2} \cdot -e^{-x} \quad \sigma'(x) = u(x)^{-1}$$

$$= (1+e^{-x})^{-2} \cdot e^{-x} \quad u'(x) = -u(x)^{-2} \cdot u'(x)$$

$$\sigma'(x) = \frac{e^{-x}}{(1+e^{-x})^2} \quad \sigma'(x) = \sigma(x)[1-\sigma(x)]$$

$$1-\sigma(x) = 1 - \frac{1}{1+e^x}$$

$$1-\sigma(x) = \frac{1+e^x}{1+e^x} - \frac{1}{1+e^x}$$

$$1-\sigma(x) = \frac{e^x}{1+e^x}$$

$$\sigma'(x) = \sigma(x)[1-\sigma(x)] \therefore$$

b) previous result + chain rule $\rightarrow$ derive gradient of the log likelihood for logistic regression
maximize negative log-likelihood

- observations: (x_i, y_i) , $y_i \in \{0, 1\}$

- logistic regression models the probability that $y_i = 1$ as: $\mu_i = \sigma(w^T x_i)$

$$\mu_i = \frac{1}{1+e^{-w^T x_i}} \quad P(y_i | x_i, w) = \mu_i^{y_i} (1-\mu_i)^{1-y_i} \quad y_i = 1 \rightarrow \mu_i$$

$$- \text{likelihood: } \mathcal{L}(w) = \prod_{i=1}^n \mu_i^{y_i} (1-\mu_i)^{1-y_i} \quad y_i = 0 \rightarrow 1-\mu_i$$

$$\log[\mathcal{L}(w)] = \sum_{i=1}^n [y_i \log(\mu_i) + (1-y_i) \log(1-\mu_i)] \quad \nabla_w \mathcal{L}(w)$$

$$\mu_i = \sigma(w^T x_i) \quad \nabla_w \mathcal{L}_i = -[y_i \nabla_w \log(\mu_i) + (1-y_i) \nabla_w \log(1-\mu_i)]$$

$$\text{Differentiate } \log(\mu_i)$$

$$\nabla_w \log(\mu_i) = \frac{1}{\mu_i} \nabla_w \mu_i$$

$$\nabla_w \mu_i \quad \mu_i = \sigma(w^T x_i) \quad \nabla_w \mu_i = \sigma'(w^T x_i) x_i$$

$$\sigma'(x) = \sigma(x)[1-\sigma(x)] \quad \sigma(w^T x_i) = \mu_i (1-\mu_i)$$

$$\nabla_w (w^T x_i) = x_i \quad \boxed{\nabla_w \mu_i = \mu_i (1-\mu_i) x_i}$$

$$\nabla_w \log(\mu_i) = \frac{1}{\mu_i} \mu_i (1-\mu_i) x_i$$

$$\boxed{\nabla_w \log(\mu_i) = (1-\mu_i) x_i}$$

$$\text{Differentiate } \log(1-\mu_i)$$

$$\nabla_w \log(1-\mu_i) = \frac{1}{1-\mu_i} \nabla_w (1-\mu_i)$$

$$\nabla_w (1-\mu_i) = -\nabla_w \mu_i$$

$$\nabla_w \log(1-\mu_i) = -\frac{1}{1-\mu_i} \nabla_w \mu_i$$

$$\nabla_w \log(1-\mu_i) = -\frac{\mu_i (1-\mu_i) x_i}{1-\mu_i}$$

$$\boxed{\nabla_w \log(1-\mu_i) = -\mu_i x_i}$$

$$\nabla_w \mathcal{L}_i = -[y_i \nabla_w \log(\mu_i) + (1-y_i) \nabla_w \log(1-\mu_i)]$$

$$\nabla_w \mathcal{L}_i = -[y_i (1-\mu_i) x_i + (1-y_i) (-\mu_i x_i)]$$

$$= -[y_i (1-\mu_i) + (1-y_i) (-\mu_i)] x_i$$

$$= -[y_i - y_i \mu_i + \mu_i - y_i \mu_i] x_i$$

$$\boxed{\nabla_w \mathcal{L}_i = (\mu_i - y_i) x_i}$$

$$\nabla_w \mathcal{L}(w) = \sum_{i=1}^n (\mu_i - y_i) x_i \quad \text{where } \mu_i = \sigma(w^T x_i)$$

$$\boxed{\nabla_w \mathcal{L}(w) = X^T (\mu - y)}$$

c) derive Hessian

$$H = X^T S X, \text{ where } S = \text{diag}(\mu_1(1-\mu_1), \dots, \mu_n(1-\mu_n))$$

Show $H \succeq 0$, H is positive semidefinite

$$\nabla_w \mathcal{L}(w) = \sum_{i=1}^n (\mu_i - y_i) x_i, \quad \mu_i = \sigma(w^T x_i)$$

$$\nabla_w \mu_i = \mu_i (1-\mu_i) x_i$$

$$H = \nabla_w^T \mathcal{L}(w) = \sum_{i=1}^n \mu_i (1-\mu_i) x_i x_i^T$$

$$\text{Let } S = \text{diag}(\mu_1(1-\mu_1), \dots, \mu_n(1-\mu_n))$$

$$\boxed{H = X^T S X}$$

to show $H \succeq 0$, let z be any vector:

$$z^T H z = z^T X^T S X z$$

$$= (X z)^T S (X z)$$

$$= \sum_{i=1}^n \mu_i (1-\mu_i) (x_i^T z)^2$$

$$\text{since } 0 < \mu_i < 1, \mu_i (1-\mu_i) \geq 0, (x_i^T z)^2 \geq 0.$$

$$\text{Hence, } z^T H z \geq 0, \text{ for all } z.$$

$$\text{Therefore, } H \succeq 0. \quad \square$$

2) derive normalization constant Z for 1-dimensional zero-mean Gaussian

$$P(x; \sigma^2) = \frac{1}{Z} \exp\left(-\frac{x^2}{2\sigma^2}\right) \text{ such that } P(x; \sigma^2) \text{ becomes a valid density.}$$

- probability density must integrate to 1 over all possible values.

$$\int_{-\infty}^{\infty} P(x; \sigma^2) dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{Z} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx = \frac{1}{Z} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx = 1$$

$$\boxed{Z = \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx}$$

$$I = \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx, \text{ then } I^2 =$$

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx \right)^2$$

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2\sigma^2} dy \right)$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} e^{-y^2/2\sigma^2} dx dy$$

$$e^a e^b = e^{a+b}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2\sigma^2} dx dy$$

because $x^2+y^2 \rightarrow$ convert to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta \quad \text{and} \quad x^2+y^2 = r^2$$

$$dx dy \rightarrow r dr d\theta$$

original x, y integral covers entire plane, so polar-coordinate bounds:

$$0 \leq r < \infty \quad \text{and} \quad 0 \leq \theta \leq 2\pi$$

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2\sigma^2} r dr d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2/2\sigma^2} dr$$

$$I^2 = 2\pi \int_0^{\infty} r e^{-r^2/2\sigma^2} dr$$

$$\int_0^{\infty} r e^{-r^2/2\sigma^2} dr$$

$$u = \frac{r^2}{2\sigma^2} = r^2 (2\sigma^2)^{-1}$$

$$du = \frac{2r}{2\sigma^2} dr = \frac{r}{\sigma^2} dr$$

$$\sigma^2 du = r dr$$

$$I^2 = 2\pi \int_0^{\infty} e^{-u} \sigma^2 du$$

$$I^2 = 2\pi \sigma^2 \int_0^{\infty} e^{-u} du$$

$$\int_0^{\infty} e^{-u} du = [-e^{-u}]_0^{\infty}$$

$$e^{-\infty} = 0, \quad e^0 = 1$$

$$= 0 - (-1) = 1$$

$$I^2 = 2\pi \sigma^2$$

$$\sqrt{I^2} = \sqrt{2\pi \sigma^2}$$

$$I = \sqrt{2\pi \sigma^2} = \sigma \sqrt{2\pi}$$

3) Regression

a) why a Gaussian prior gives ridge regression

- linear regression model w/ zero-mean Gaussian prior on weights:

$$P(w) = \prod_{i=1}^n \mathcal{N}(w_i | 0, \tau^2)$$

- then MAP problem = ridge regression:

$$\arg \max_w \left[\sum_{i=1}^n \log \mathcal{N}(y_i | w_0 + w^T x_i, \sigma^2) \right] + \sum_{i=1}^n \log \mathcal{N}(w_j | 0, \tau^2) = \arg \max_w \left[\sum_{i=1}^n (y_i - (w_0 + w^T x_i))^2 + \lambda \|w\|_2^2 \right], \quad \lambda = \frac{\sigma^2}{\tau^2}$$

- Gaussian density

$$z \sim \mathcal{N}(\mu, \sigma^2), \quad \mathcal{N}(z | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right)$$

$$\log \mathcal{N}(z | \mu, \sigma^2) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2}$$

$$\sum_{i=1}^n \log \mathcal{N}(y_i | w_0 + w^T x_i, \sigma^2) + \sum_{j=1}^p \log \mathcal{N}(w_j | 0, \tau^2)$$

$$= C - \frac{1}{2\sigma^2} \sum_{i=1}^n [y_i - (w_0 + w^T x_i)]^2 - \frac{1}{2\tau^2} \sum_{j=1}^p w_j^2$$

where C contains terms independent of w since: $\sum_{j=1}^p w_j^2 = \|w\|_2^2$

and maximizing the MAP objective is equivalent to minimizing:

$$\frac{1}{2\sigma^2} \sum_{i=1}^n [y_i - (w_0 + w^T x_i)]^2 + \frac{1}{2\tau^2} \|w\|_2^2$$

Dropping the positive multiplicative constants and using the normalization convention given creates the ridge objective:

$$\frac{1}{N} \sum_{i=1}^n [y_i - (w_0 + w^T x_i)]^2 + \lambda \|w\|_2^2, \quad \text{with } \lambda = \frac{\sigma^2}{\tau^2}$$

b) closed form solution x^* to ridge regression problem: minimize $\|Ax - b\|_2^2 + \lambda \|x\|_2^2$

- find x^* that minimizes $f(x)$: closed objective $\rightarrow$ take gradient $\rightarrow$ set gradient $= 0 \rightarrow$ solve for x

$$f(x) = \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

$$= (Ax - b)^T (Ax - b) + \lambda x^T x$$

$$= A^T A x - 2A^T b + b^T b + \lambda x^T x$$

$$\nabla_x f = 2A^T A x - 2A^T b + 2\lambda \Gamma^T x$$

- @ minimize x^* , set gradient $= 0$

$$2A^T A x^* - 2A^T b + 2\lambda \Gamma^T x^* = 0$$

$$A^T A x^* + \Gamma^T x^* = A^T b$$

$$x^* (A^T A + \Gamma^T \Gamma) = A^T b$$

- assuming $A^T A + \Gamma^T \Gamma$ is invertible:

$$x^* = (A^T A + \Gamma^T \Gamma)^{-1} A^T b$$

- for ridge regression: $\Gamma^2 = \lambda I$ gives $\Gamma^T \Gamma = \lambda I$

- therefore,

$$\boxed{x^* = (A^T A + \lambda I)^{-1} A^T b}$$

d) change model from putting bias inside feature vector to writing it separately:

$$\hat{y} = \theta^T x + b \quad \text{minimize } \|Ax + b\mathbf{1} - y\|_2^2 + \lambda \|x\|_2^2$$

- solve for x^* explicitly

$$f(x, b) = \|Ax + b\mathbf{1} - y\|_2^2 + \lambda \|x\|_2^2, \text{ where } A = \text{feature matrix}, b = \text{bias}, x = \text{weight vector}, \mathbf{1} = \text{column vector of 1s}$$

- first solve for optimal bias: b^*

$$\frac{df}{db} \left[\|Ax + b\mathbf{1} - y\|_2^2 + \lambda \|x\|_2^2 \right] = 2\mathbf{1}^T (Ax + b\mathbf{1} - y) = 0$$

$$0 = 2\mathbf{1}^T (Ax + b\mathbf{1} - y)$$

$$= 2\mathbf{1}^T (Ax + b\mathbf{1}) - 2\mathbf{1}^T y$$

$$= 2\mathbf{1}^T Ax + 2b\mathbf{1}^T \mathbf{1} - 2\mathbf{1}^T y$$

- say we can observe m observations of $2\mathbf{1}^T \mathbf{1} = m$

$$= \mathbf{1}^T Ax + mb - \mathbf{1}^T y$$

$$mb = \mathbf{1}^T y - \mathbf{1}^T Ax$$

$$mb = \mathbf{1}^T (y - Ax)$$

$$\boxed{b^* = \frac{1}{m} \mathbf{1}^T (y - Ax)}$$

$$\text{or } b^* = \bar{y} - \bar{a}^T x, \text{ where } \bar{a} \text{ is the vector of feature means}$$

- now solve for x^* by taking the gradient w/ respect to x

$$\nabla_x f = 2A^T (Ax + b\mathbf{1} - y) + 2\lambda \Gamma^T x$$

$$0 = 2A^T (Ax + b\mathbf{1} - y) + 2\lambda \Gamma^T x$$

$$0 = A^T (Ax + b\mathbf{1} - y) + \lambda \Gamma^T x$$

$$= A^T Ax + A^T b\mathbf{1} - A^T y + \lambda \Gamma^T x$$

$$= A^T b\mathbf{1} - A^T y + A^T A x + \lambda \Gamma^T x$$

$$A^T y = A^T b\mathbf{1} = x^* (A^T A + \Gamma^T \Gamma)$$

$$x^* = A^T (y - b\mathbf{1}) (A^T A + \Gamma^T \Gamma)^{-1}$$

$$b^* = \frac{1}{m} \mathbf{1}^T (y - Ax) \quad C = I - \frac{1}{m} \mathbf{1} \mathbf{1}^T$$

$$Ax + b\mathbf{1} - y = C(Ax - y)$$

optimization becomes $\min_x \|C(Ax - y)\|_2^2 + \lambda \|x\|_2^2 \rightarrow$ same as 3a)

$$0 = 2A^T C(Ax - y) + 2\lambda \Gamma^T x$$

$$0 = A^T CAx - A^T Cy + \lambda \Gamma^T x$$

$$A^T Cy = x^* (A^T AC + \Gamma^T \Gamma)$$

$$x^* = (A^T AC + \Gamma^T \Gamma)^{-1} A^T Cy, \text{ where } C = I - \frac{1}{m} \mathbf{1} \mathbf{1}^T$$

once x^* is known:

$$b^* = \frac{1}{m} \mathbf{1}^T (y - Ax^*)$$

- specialize to ridge regression: $\Gamma^T \Gamma = \lambda I$

$$\boxed{x^* = (A^T CA + \lambda I)^{-1} A^T Cy}$$

$$b^* = \frac{1}{m} \mathbf{1}^T (y - Ax^*)$$

e) $f(w, b) = \|XW + b\mathbf{1} - y\|_2^2 + \lambda \|W\|_2^2$

$$\nabla_w f = 2X^T (XW + b\mathbf{1} - y) + 2\lambda \Gamma^T W$$

$$\frac{df}{db} = 2\mathbf{1}^T (XW + b\mathbf{1} - y)$$

$$\frac{df}{db} = 2\mathbf{1}^T (XW + b\mathbf{1} - y)$$

$$\text{ridge regression: } \Gamma^T \Gamma = \lambda I$$

$$\nabla_w f = 2X^T (XW + b\mathbf{1} - y) + 2\lambda W$$

$$\frac{df}{db} = 2\mathbf{1}^T (XW + b\mathbf{1} - y)$$

$$\text{gradient descent: } w^{(t+1)} = w^{(t)} - \eta \nabla_w f$$

$$b^{(t+1)} = b^{(t)} - \eta \frac{df}{db}$$